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“Custom Flight Controller Development for UAV Applications”



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Project Proposal

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1. Introduction

1.1 Background and societal Need

The growing integration of autonomous and semi-autonomous systems across industrial, commercial, and research sectors has elevated the importance of advanced embedded control platforms. Modern applications in infrastructure inspection, environmental monitoring, public safety, logistics, agriculture, and robotics rely on precise, real-time sensor processing and stable control algorithms to ensure safe and efficient operation. As industries accelerate toward automation, the demand for engineers capable of designing, analyzing, and implementing control systems at the hardware and firmware levels continues to increase.

Many existing flight-control solutions, while mature, are designed around fixed configurations or proprietary architectures that limit deep system customization, scalability, or adaptation to specialized mission profiles. Industrial applications often require hardware capable of supporting heterogeneous sensors, redundant configurations, and flexible communication interfaces to accommodate varying operational standards and safety requirements. Furthermore, as autonomous systems evolve, industries increasingly favor platforms that offer transparency, long-term maintainability, and the ability to integrate custom logic or specialized control algorithms without dependence on closed ecosystems.

These trends highlight a clear need for modular, adaptable, and technically open flight-control architectures that can be tailored to different UAV platforms and operational scenarios. A custom controller designed with modern performance requirements, robust interfacing capabilities, and compatibility with established firmware ecosystems directly addresses this gap, supporting both current industry practices and anticipated future developments in autonomous aerial systems.

1.2 Project overview and Objective

This project proposes the design and development of a fully custom flight-controller platform intended for use across multiple unmanned aerial vehicle (UAV) configurations. The controller is built around a high-performance STM32H743 microcontroller operating at 480 MHz, selected for its computational capacity, extensive peripheral set, and suitability for real-time control tasks. The hardware integrates a complete suite of essential flight sensors, including inertial measurement units, barometric pressure sensing, magnetometers, and satellite-based positioning modules. These components enable accurate attitude estimation, altitude determination, heading reference, and global navigation, forming the sensor backbone required for stable flight and autonomous operation.

A key design goal is modularity. The board incorporates standard communication interfaces allowing integration with diverse UAV frames, ESC systems, GPS receivers, radio-control protocols, and auxiliary sensors. The hardware architecture is engineered to maintain compatibility with leading open-source firmware ecosystems such as ArduPilot, iNav, and Betaflight, ensuring interoperability while also enabling the development of custom firmware tailored to advanced applications. This dual-purpose approach supports both rapid validations using established software and deeper experimentation through original embedded-systems development.

Physical design adheres to practical constraints, targeting a compact form factor suitable for 50 mm flight-controller mounting standards commonly used in quadcopters and adaptable to fixed-wing VTOL platforms. The project includes full PCB schematic design, signal-integrity considerations, component selection, power-distribution architecture, and manufacturing through a budget-conscious prototyping process. By June, the team will demonstrate stable manual and autonomous quadcopter flight as proof of concept, validating both the hardware platform and the foundational firmware components.

The project showcases a complete engineering workflow from concept and design to implementation and testing while delivering a functional controller that is cost-efficient, educationally valuable, and aligned with emerging industry needs for adaptable autonomous-system hardware.

1.3 Project Objectives

The objective of this project is to design and implement a custom, modular flight-controller hardware platform that can support a broad range of UAV firmware ecosystems and vehicle configurations. The primary focus is the development of robust electronic architecture; firmware development is limited to essential board validation and flight testing.

The project aims to achieve the following:

1. Develop a complete custom PCB flight controller, integrating a high-performance STM32 microcontroller, essential flight sensors, GNSS support, and a reliable power-distribution and protection system.
2. Implement a modular hardware interface layout, providing standardized UART, SPI, I²C, PWM, USB, and power connections to ensure compatibility with diverse sensors, ESCs, radio systems, and potential future expansions.

3. Align the hardware design with established open-source firmware ecosystems, enabling the board to operate with ArduPilot, iNav, Betaflight, or other platforms without requiring significant architectural changes.
4. Perform minimal required firmware bring-up, limited to sensor communication tests, bus verification, and basic flight-stability checks needed to validate hardware functionality.
5. Demonstrate stable quadcopter flight, using either existing firmware or simplified custom implementation, solely to verify the electrical and architectural integrity of the hardware platform.

1.4 Project Scope

This project encompasses the design and implementation of a custom, modular flight-controller hardware platform together with the development of essential flight-control algorithms required for stable manual flight and basic autonomous operation. The scope includes complete PCB and hardware architecture design, sensor integration, power-management circuitry, and communication-interface layout. On the firmware side, the project will implement core stabilization logic, cascaded PID control loops, sensor fusion for attitude estimation, and simple autonomous behaviors such as basic waypoint navigation or position hold. Advanced mission planning, complex autonomy, and large-scale firmware ecosystems remain outside the project's scope, as the primary objective is to deliver a functional and extensible hardware platform validated through controlled, stable flight.

1.5 Limitations

Autonomous functionality is restricted to basic behaviors intended for validation purposes rather than operational deployment. Environmental robustness, regulatory certification, and extended reliability testing fall outside the scope of this work and are assumed to be addressed in future iterations. It is further assumed that standard UAV components such as motors, ESCs, batteries, and radio-control systems operate within nominal specifications and are compatible with the designed hardware platform. Moreover, the initial design will be based on industrial standards and will follow codes and protocols that take certification requirements from (EASA , FAA) into account.

2. Technical Approach and Methodology

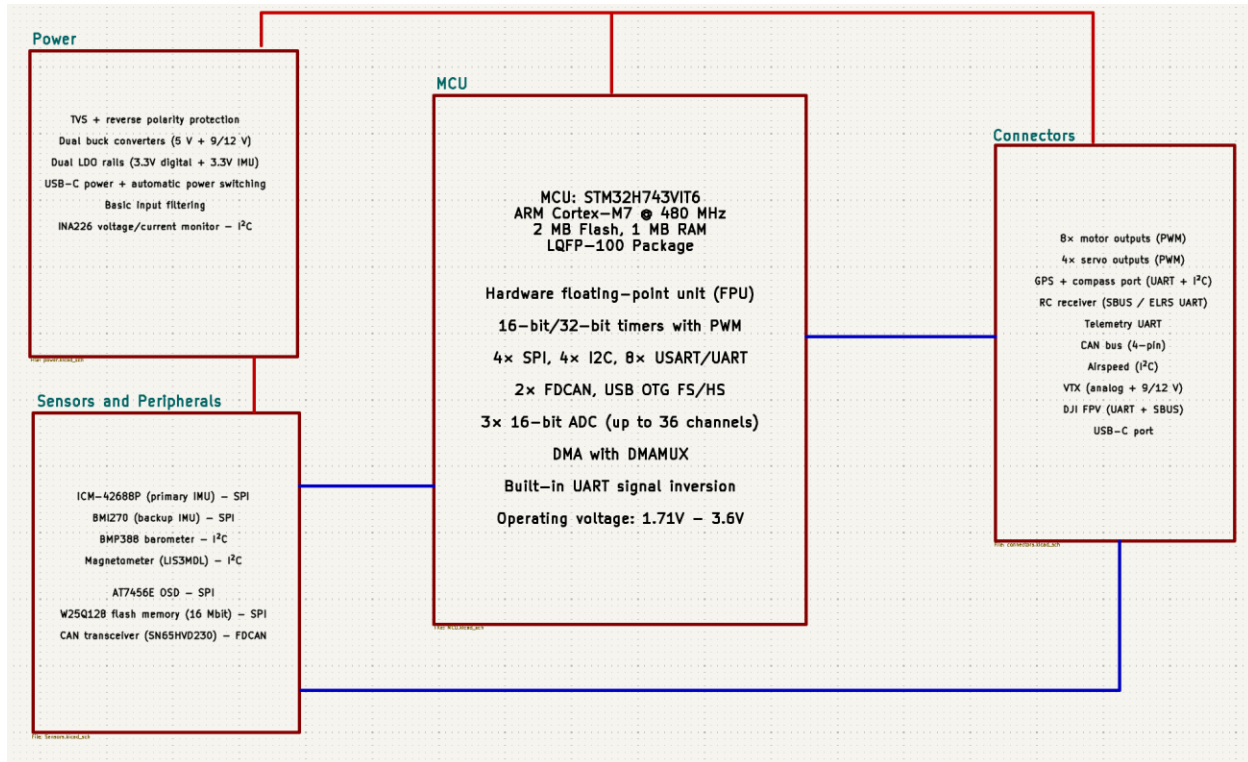


Figure [1]

2.1 Overall System Architecture

The technical approach of this project is centered on the development of modular, hardware-first flight-controller architecture capable of supporting multiple UAV configurations and firmware ecosystems. The system is organized into four primary subsystems: power management, processing and control, sensing and peripheral integration, and external connectivity. A block-level representation of the complete architecture is shown in **Figure 1**, illustrating the relationship between the microcontroller core and the surrounding hardware modules.

At the center of the system is a high-performance STM32H743 microcontroller, selected for its real-time processing capability, extensive peripheral support, and suitability for flight-control applications. Surrounding the MCU, dedicated power-conditioning circuitry ensures stable operation under electrically noisy UAV environments, while multiple sensor interfaces and communication ports enable flexible integration with external components such as ESCs, GPS modules, radio receivers, and telemetry systems.

The architecture prioritizes hardware modularity and electrical robustness. Standardized interfaces (SPI, I²C, UART, PWM, CAN, USB) are exposed through dedicated connectors, allowing the flight controller to operate with different firmware platforms and airframe types without requiring hardware redesign. Firmware development is treated as a validation layer, used primarily to confirm sensor communication, control-loop execution, and overall system stability.

2.2 Power Management System

The power-management and protection subsystem is designed to ensure stable and reliable operation under the electrical conditions typical of UAV platforms, including voltage fluctuations, high-current transients, and electromagnetic interference generated by motors and ESCs. Reverse-polarity protection, transient voltage suppression, and input filtering are implemented at the power entry to protect against incorrect connection and fast electrical disturbances. It is worth noting that there will be a dedicated power distribution board for the motors, servos and power demanding modules. Therefore, expanding the compatibility borders and reinforcing the separation of low-power and high-power systems.

Voltage regulation is achieved through high-efficiency buck converters generating intermediate supply rails, followed by dedicated low-dropout regulators supplying isolated 3.3 V domains for digital logic and inertial sensors. This staged regulation approach reduces noise coupling between high-current digital loads and noise-sensitive sensing circuitry. Voltage and current monitoring are integrated via an I²C-connected sensing device to support diagnostics and power-awareness. A USB-C power interface with automatic source selection is included to support development and testing without reliance on the main battery.

2.2.1 Design Standards and Protocols

The power subsystem is designed in accordance with established industrial best practices and standards. Electromagnetic immunity and transient robustness follow the principles of the **IEC 61000** series. Low-voltage safety considerations are guided by **IEC 62368-1**. PCB layout and power-distribution practices adhere to **IPC-2221** and **IPC-9592** recommendations, while USB-C power behavior aligns with the USB Type-C specification. These standards are applied as design references rather than certification targets.

2.3 Processing and Control Core

The processing and control core of the flight controller is based on the STM32H743 microcontroller, featuring an ARM Cortex-M7 core operating at up to 480 MHz with single-precision floating-point support (FPU). This processing capability is sufficient to execute real-time attitude estimation, cascaded PID control loops, and peripheral communication within deterministic timing constraints typical of UAV flight-control systems.

The microcontroller provides 1 MB of on-chip RAM and 2 MB of Flash memory, allowing separation of program memory, runtime data, and DMA buffers. Multiple Advanced Control Timers (TIM1/TIM8) and general-purpose timers support generation of motor and servo control signals with microsecond-level resolution, enabling precise PWM outputs for multicopter and fixed-wing actuation. Timer allocation is selected to maintain independent control of motor outputs and auxiliary actuators.

The clock system uses an external high-speed crystal oscillator (HSE) as the primary reference source to ensure stable timing for control loops and communication peripherals. The internal PLL is configured to achieve the target core frequency while maintaining peripheral clock domains within their specified limits. This clocking approach improves timing determinism compared to internal RC oscillators, particularly for high-speed SPI communication and control-loop scheduling.

Power is supplied to the MCU through multiple dedicated supply pins (VDD, VDDA, VDDUSB), each locally decoupled using a combination of bulk capacitors (μF range) and high-frequency ceramic capacitors (100nF and below). Analog and digital supply domains are treated separately at the schematic level to reduce coupling of digital switching noise into ADCs and other noise-sensitive circuits.

The STM32H743 provides multiple SPI, I²C, UART/USART, ADC, USB 2.0 FS, and CAN-FD capable interfaces, enabling clean separation of sensor, control, and communication traffic. High-bandwidth sensors such as IMUs are connected via SPI, while lower-speed peripherals are assigned to I²C or UART interfaces. Direct Memory Access (DMA) is employed for SPI and UART data transfers to reduce CPU load and ensure predictable execution of time-critical control tasks.

Pin multiplexing and peripheral mapping are selected to avoid resource conflicts and to maintain compatibility with established open-source firmware platforms. Debug and programming access is provided through SWD, allowing low-level firmware development, register-level debugging, and performance analysis during system bring-up and validation.

2.3.1 Design Standards and Protocols

The processing and control core is designed following established embedded-system and PCB design practices addressing signal integrity, electromagnetic compatibility, and timing stability. Clock distribution, power decoupling, and grounding strategies are implemented in accordance with **IPC-2221** recommendations and EMC-oriented design principles derived from the **IEC 61000** framework. USB connectivity associated with the microcontroller follows the **USB 2.0** electrical specification to ensure reliable communication and signal integrity

2.4 Sensors and Peripherals

The sensor and peripheral subsystem are responsible for acquiring all primary state variables required for flight stabilization, navigation, and pilot feedback. The design integrates both onboard sensors and standardized external interfaces to support multirotor, fixed-wing, and VTOL configurations while maintaining compatibility with existing UAV firmware ecosystems.

2.4.1 Inertial Measurement Units (IMUs)

The flight controller integrates two independent 6-axis inertial measurement units:

- Primary IMU: InvenSense ICM-42688-P
- Secondary IMU: Bosch BMI270

Both devices provide tri-axis accelerometer and gyroscope measurements and operate from a 3.3 V $\pm 5\%$ supply. The IMUs are interfaced via SPI, selected over I²C to meet deterministic latency and bandwidth requirements for control loops operating at ≥ 1 kHz.

Each IMU is connected to the MCU via:

- Dedicated chip-select (CS) line
- Shared SPI clock (SCK), MOSI, MISO

The SPI bus is configured to support clock rates up to 8–10 MHz, well within the electrical limits of both devices, ensuring sufficient bandwidth for:

- Gyroscope output data rates up to 4 kHz (ICM-42688-P)
- Accelerometer output data rates up to 1.6 kHz

This configuration enables raw inertial data acquisition with sub-millisecond latency. Redundant IMU integration allows:

- Sensor cross-validation
- Firmware-level failover
- Improved robustness under vibration and transient disturbances

2.4.2 Barometric and Air-Pressure Sensing

Altitude estimation is provided by an onboard BMP388 barometric pressure sensor, interfaced via I²C. The sensor operates from a 3.3 V supply and supports pressure resolution on the order of ± 0.08 m under nominal conditions, sufficient for altitude-hold control and vertical velocity estimation.

The I²C bus operates at 400 kHz (Fast mode), balancing update rate and bus stability. Pull-up resistors are selected to meet I²C rise-time requirements for the expected bus capacitance, including external devices.

In addition to onboard barometric sensing, a dedicated external airspeed/pressure connector is provided for differential pressure sensors. This interface supplies:

- Regulated 3.3 V power
- I²C communication

This enables air speed measurement for fixed-wing and VTOL platforms without additional analog front-end circuitry on the flight controller.

2.4.3 GPS and Magnetometer Interface

Global position and heading reference are provided through a combined GPS and magnetometer module connected via a dedicated external connector. The interface supplies regulated power and digital communication lines required for position, velocity, and heading estimation.

The GPS receiver communicates with the flight controller through a UART interface operating at 3.3 V logic levels, supporting baud rates up to 115200 bps. This bandwidth is sufficient for standard GPS message formats (e.g., NMEA sentences or vendor-specific binary protocols) at update rates commonly ranging from 1 Hz to 10 Hz. The UART interface is electrically isolated from high-bandwidth sensor buses to prevent interference with time-critical inertial data acquisition.

The integrated magnetometer within the GPS module is interfaced via I²C, operating at 3.3 V and configured for 400 kHz fast-mode operation. Locating the magnetometer within an external GPS module allows physical separation from high-current conductors, voltage regulators, and switching elements on the flight controller, reducing susceptibility to magnetic interference without requiring onboard shielding or compensation circuitry.

The GPS/magnetometer connector provides:

- Regulated supply voltage (5V)
- UART TX/RX lines for GPS data
- I²C SCL/SDA lines for magnetometer communication
- Ground reference

This connector-level definition enables direct compatibility with widely used UAV GPS–compass modules and allows the same flight controller hardware to be deployed across different airframes without modification.

2.4.4 On-Screen Display and Digital Video Systems

An onboard AT7456E OSD controller is interfaced via SPI to support real-time telemetry overlay on analog video signals. The SPI interface is operated within the device’s specified clock limits to ensure stable character rendering and video synchronization.

In parallel, the flight controller includes a dedicated connector for DJI Air Unit–class digital video systems, providing:

- Regulated power rails (9 V / 12 V, as required)
- UART communication for telemetry and configuration
- Control signaling compatible with DJI digital FPV ecosystems

Electrical pin assignments and voltage levels are selected to match DJI Air Unit interface requirements, enabling direct integration without external adapters.

2.4.5 CAN Bus Interface and Peripheral Support

The flight controller integrates a dedicated CAN transceiver connected to the MCU’s FDCAN peripheral, enabling robust differential communication with external UAV subsystems. The CAN interface is designed to support distributed peripherals such as smart ESCs, power modules, range sensors, or companion control boards commonly used in modern UAV architectures.

The CAN transceiver operates at 3.3 V logic levels on the MCU side and interfaces to a differential CAN bus (CANH/CANL) on the external side. The design supports standard CAN physical-layer requirements, including differential signaling and common-mode noise rejection, making it suitable for electrically noisy environments typical of propulsion and power systems.

A dedicated CAN connector is provided to expose the CAN bus externally, supplying:

- CANH and CANL differential signal lines
- Ground reference
- Optional power rail (depending on peripheral requirements)

The physical-layer implementation is compatible with standard CAN bus topologies and supports configurable bit rates within the operating limits of the transceiver and MCU FDCAN peripheral. Bus termination is treated as a system-level consideration and is expected to be applied externally according to the specific network topology and connected devices.

By integrating the CAN transceiver onboard while exposing the bus through a standardized connector, the design enables modular expansion toward CAN-based peripherals without imposing fixed assumptions about network size or device roles.

2.4.6 Airspeed Sensor Interface

The flight controller provides a dedicated airspeed sensor connector intended for integration of external differential air-pressure sensors commonly used in fixed-wing and VTOL configurations. The interface is defined at the electrical and protocol level to allow compatibility with a range of commercially available airspeed modules without constraining the specific sensor implementation.

The airspeed connector provides:

- Regulated 3.3 V supply
- I²C communication interface (SCL, SDA)
- Ground reference

The I²C interface supports standard-mode (100 kHz) and fast-mode (400 kHz) operation, consistent with common digital airspeed sensors. Pull-up resistors are implemented on the flight controller to satisfy I²C bus electrical requirements, accounting for external sensor modules and cable length.

By defining the airspeed interface as a digital I²C connection rather than an analog input, the design avoids the need for onboard analog signal conditioning while maintaining flexibility in sensor selection.

2.4.7 Design Standards and Protocols

The sensor and peripheral subsystem are designed in accordance with established digital communication and interface standards. SPI-based devices follow the Motorola SPI specification, while I²C-connected sensors adhere to the NXP I²C-bus specification, including open-drain signaling and pull-up requirements. Asynchronous serial communication for GPS, telemetry, and digital video control interfaces follow UART framing conventions. The CAN interface is implemented in accordance with **ISO 11898-1** and **ISO 11898-2**, defining the CAN data link and physical layers. Analog video signaling, and airspeed sensor for the onboard OSD follows standard composite video conventions. Connector definition and schematic-level interface design adhere to **IPC-2221** guidelines. Layout-dependent EMC and signal-integrity considerations are addressed separately in the PCB layout section.

2.5 External Interfaces and Connector Architecture

The external interface architecture defines the modular expansion capability of the flight controller and establishes clear electrical boundaries between the onboard core subsystems and user-selectable external modules. While critical sensing and processing functions are implemented through fixed onboard components, all mission-dependent peripherals actuation, communication, navigation modules, and video systems are interfaced through dedicated connectors with explicitly defined electrical contracts.

This separation allows the same hardware platform to support multiple UAV configurations without schematic modification, while preserving deterministic behavior for time-critical onboard subsystems.

2.5.1 Motor and Servo Control Interfaces

The flight controller provides a total of 12 independent PWM output channels, all configurable at the firmware level and mapped to MCU timer resources. This output capacity is selected to support a wide range of UAV configurations, including multirotor, fixed-wing, and VTOL platforms, without requiring hardware modification.

The standard allocation of PWM outputs is organized as follows:

- 4 PWM outputs dedicated to a 4-in-1 ESC interface, intended for compact multirotor configurations using integrated ESC modules
- 4 PWM outputs exposed as individual motor control pins, supporting discrete ESCs commonly used in larger multirotor or hybrid configurations
- 4 PWM outputs allocated to servo control, enabling actuation of control surfaces, tilt mechanisms, or auxiliary devices in fixed-wing and VTOL systems

All PWM outputs operate at 3.3 V logic levels and are generated using hardware timers to ensure deterministic timing and consistent pulse width generation. Output frequency and pulse width are configurable through firmware to support standard ESC protocols and servo control requirements.

Servo power is not sourced from the flight controller logic supply. Instead, servo power is provided externally through the system stack or a dedicated power-distribution board. This design choice allows the use of high-voltage servos and prevents high-current or high-voltage actuation loads from coupling into sensitive onboard electronics. The flight controller therefore supplies only the control signal and ground reference for servo outputs, while power delivery is managed externally according to application requirements.

This separation of control and power domains improves electrical robustness, increases compatibility with a wider range of actuators, and aligns with modular UAV system architectures.

2.5.2 Radio Control Receiver Interface

A dedicated RC receiver connector supports serial receiver protocols commonly used in UAV systems, including SBUS and CRSF (ELRS). The interface provides:

- Regulated supply voltage
- UART signal input at 3.3 V logic levels
- Ground reference

The receiver interface is electrically isolated from high-bandwidth sensor buses and motor outputs, ensuring reliable reception of control data under electrically noisy operating conditions.

2.5.3 Telemetry

A dedicated telemetry UART interface is exposed through connector pins to support bidirectional data exchange with external ground stations, radios, or companion computers. This interface operates at 3.3 V logic levels and supports configurable baud rates within the UART capabilities of the MCU.

For development and system bring-up, a standard ST-Link debug connector is provided, exposing SWD signals, power reference, and ground. This interface enables low-level firmware flashing, debugging, and register-level inspection without interfering with operational connectors.

2.5.4 Video and OSD Interfaces Telemetry

The flight controller supports both analog and digital video systems. An analog FPV connector provides video signal routing and power for traditional analog camera and VTX configurations, interfacing directly with the onboard OSD controller.

For digital video systems, a dedicated DJI Air Unit connector is implemented. This interface provides:

- Regulated power rail suitable for DJI digital video units
- UART communication for telemetry and configuration
- Control signals required for full integration with DJI FPV ecosystems

The connector pinout is designed to support multiple DJI Air Unit variants, enabling direct plug-and-play integration without external adapter boards.

Some of these variants are:

1. DJI O3 Air Unit
2. DJI FPV Air Unit v1
3. DJI FPV Vista / Caddx Vista
4. DJI RunCam Link
5. DJI Air Unit Lite (Vista variants)



DJI O3 Air Unit



DJI FPV Air Unit v1

2.5.5 Connector Reuse and User Responsibility

Several external connectors on the flight controller are defined by protocol and electrical characteristics rather than fixed functional roles. If the electrical constraints and communication protocol requirements of a given connector are satisfied, certain interfaces may be repurposed to support alternative peripherals. Examples include reassigning UART-based connectors for telemetry, control, or peripheral communication, or using I²C connectors for different compatible sensor modules.

This design approach increases hardware flexibility while preserving deterministic electrical behavior. Responsibility for ensuring protocol compatibility, voltage requirements, and correct peripheral configuration rests with the user when repurposing connectors beyond their typical application. The flight controller hardware guarantees defined voltage levels, signal integrity at the schematic level, and protocol-compliant electrical interfaces, but does not enforce functional constraints on externally connected devices.

This separation of hardware capability and application-level responsibility allows advanced users to adapt the platform to specialized use cases without requiring hardware modification.

2.6 PCB Layout, Stack-Up, Signal Integrity, and EMC Design

The PCB layout and stack-up are treated as a primary design axis due to their direct impact on electromagnetic emissions, susceptibility, and overall system reliability. Given the mixed-signal nature of the flight controller combining high-speed digital interfaces, switching power supplies, sensor electronics, and high-current actuation paths, the layout is explicitly designed to meet EMI/EMC design-for-compliance requirements, rather than relying on post-layout mitigation.

2.6.1 Six-Layer Stack-Up Architecture

A six-layer PCB stack-up is employed to provide controlled return paths, effective electromagnetic shielding, and low-impedance power distribution. The layer assignment is organized as follows:

- Top layer: Signal routing referenced to an adjacent solid ground plane
- Layer 2: Power plane (VCC)
- Layer 3: Solid signal ground reference
- Layer 4: Segregated power plane for noisy or high-current domains
- Layer 5: Solid ground plane
- Bottom layer: Signal routing referenced to an adjacent ground plane

This ground-power-ground sandwich structure ensures that all high-edge-rate signals are routed adjacent to continuous reference planes, minimizing loop area and reducing both radiated and conducted emissions.

2.6.2 EMI-Driven Grounding and Return-Path Control

Electromagnetic emissions are primarily controlled through return-path integrity rather than component-level filtering. Continuous ground planes are maintained across internal layers to provide predictable, low impedance return paths for all signal currents. Signal routing is constrained such that return currents remain directly beneath their corresponding signal traces, avoiding plane discontinuities and unintended current loops.

Split grounds are intentionally avoided. Instead, noise isolation between sensitive sensor circuits and high-energy digital or power domains is achieved through physical separation, controlled routing, and plane referencing, which is consistent with modern EMC design practice.

2.6.3 Power Plane Segregation and Noise Containment

Power planes are separated by function to limit the propagation of switching noise across the PCB. High-current and high-di/dt power paths are confined to dedicated internal layers and routed to minimize loop area. This approach reduces common-mode noise injection into signal reference planes and external cables, a common source of EMI failures.

Decoupling and bulk capacitors are placed to form localized current loops between power and ground planes, further reducing high-frequency emissions.

2.6.4 EMC and Certification-Oriented Design Standards

The PCB layout and stack-up are designed in alignment with recognized EMC standards used in electronic product certification:

- **IEC 61000-6-3 / IEC 61000-6-4** — Emission standards for residential, commercial, and industrial environments
- **IEC 61000-6-1 / IEC 61000-6-2** — Immunity standards for residential and industrial environments
- **IEC 61000-4-2 / -4 / -5** — ESD, EFT, and surge immunity test references (design-for-compliance)
- **CISPR 32** — Radiated and conducted emissions for multimedia and digital equipment
- **IPC-2221** — PCB layout, grounding, and stack-up design guidelines

These standards guide layout decisions related to ground continuity, return-path control, plane usage, and connector placement. While formal EMC certification testing is outside the scope of this project, the layout is explicitly designed to maximize the likelihood of first-pass compliance under standard EMI/EMC test conditions.

3. Project Timeline and Task Allocation

3.1 Project Timeline

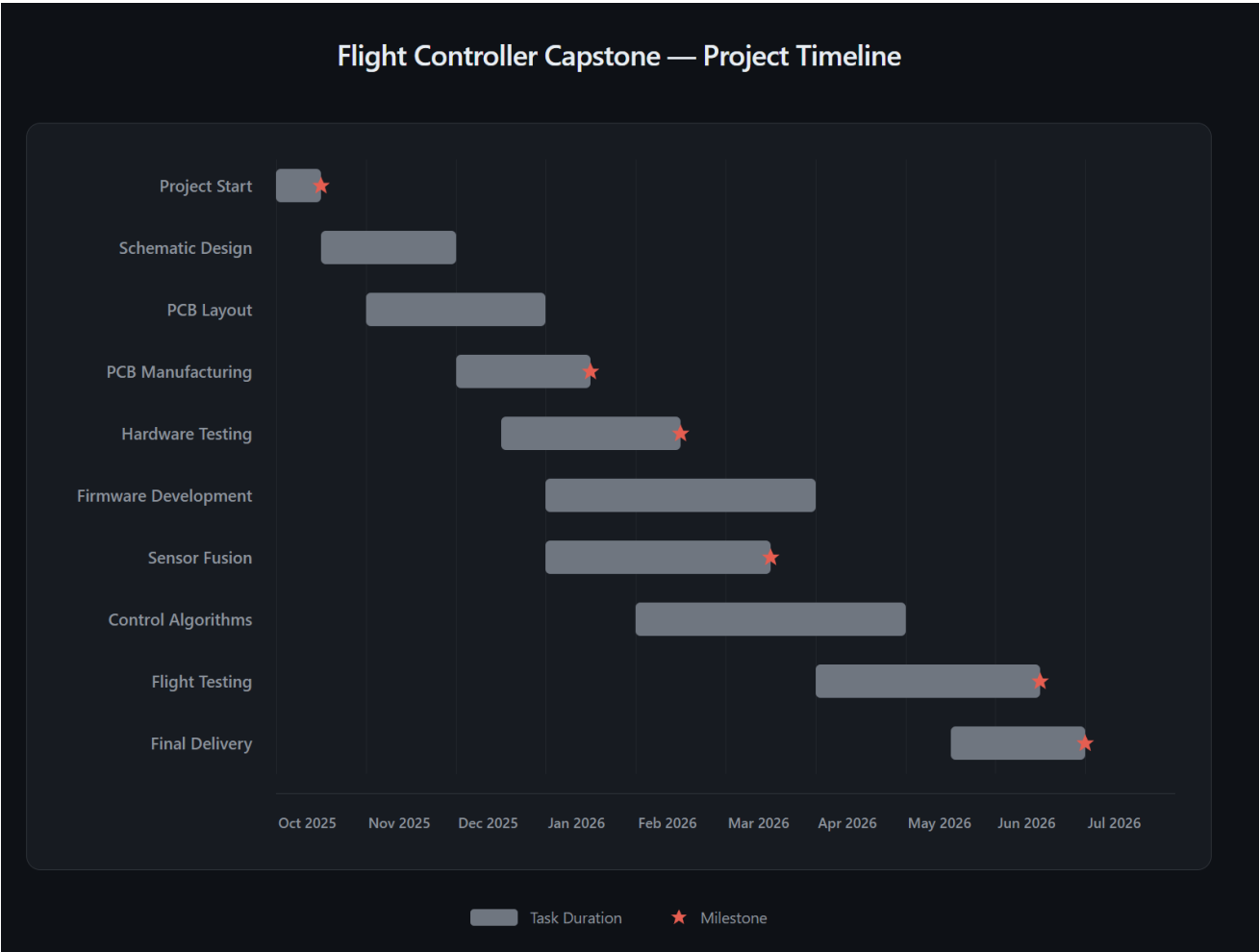


Figure [2]

Figure 2 presents the project timeline from October 2025 to June 2026, illustrating the sequencing, overlap, and key milestones of the development process. The timeline reflects a hardware-first approach, ensuring early validation of critical design decisions while enabling parallel firmware and algorithm development.

The project begins in October 2025 with initial planning and schematic design. PCB layout and stack-up design followed in November 2025, leading to PCB manufacturing in December 2025. A key milestone marks the availability of an initial test board, intended for electrical validation and bring-up rather than final deployment.

From December 2025 to January 2026, hardware testing and bring-up are conducted, enabling firmware development to commence in January 2026. Firmware development continues through March 2026, overlapping with sensor fusion and filtering, which culminates in a milestone indicating validated IMU-based state estimation.

Control algorithm development is carried out between February and April 2026, focusing on stabilization and manual flight control. System integration and flight testing take place from April to May 2026, with a milestone marking successful initial flight validation. The final phase, extending into June 2026, is dedicated to documentation, final validation, and project delivery.

This structured timeline ensures efficient use of development time, risk reduction through early hardware validation, and sufficient margin for integration and testing prior to final submission.

3.2 Task Allocation

The project is carried out by a multidisciplinary team with responsibilities allocated according to technical specialization while maintaining sufficient overlap to ensure integration and continuity. Task distribution is structured to support a hardware-first development approach while enabling parallel progress in firmware and system integration

1. **Akaki Gvelesiani:** Hardware Design, PCB Layout, and Low-Level Firmware Development responsibilities include system architecture definition, schematic design, six-layer PCB layout and stack-up development, power distribution, sensor and interface integration, and EMI/EMC-aware layout practices. In addition to hardware responsibilities, this role focuses on low-level firmware, including microcontroller bring-up, clock and reset configuration, hardware abstraction layer setup, peripheral initialization, sensor fusion, filtering, and driver development for sensors, communication interfaces, and timers. Close interaction between hardware and low-level firmware ensures rapid validation during bring-up and early testing.
2. **Saleh Alhomeidy:** Hardware Design and High-Level Firmware. Development responsibilities include shared ownership of schematic design and PCB layout, design review of routing, grounding, and stack-up decisions, and coordination of hardware revisions. This role additionally focuses on high-level firmware, including state estimation, control logic implementation, and tuning of stabilization algorithms for manual flight. By combining hardware insight with control-level firmware development, this role ensures that algorithm design is aligned with real hardware constraints and sensor behavior.

3. **Levani Kazaishvili:** Component Sourcing, Budget Management, and Mechanical Integration: include component selection and availability verification, procurement and cost tracking, budget management, and consideration of mechanical constraints such as board dimensions, mounting patterns, connector accessibility, and integration with airframe and stack components. This role supports design decisions by ensuring feasibility with respect to supply chain constraints, cost limitations, and physical integration requirements.

3.3 Test Platform

System validation and flight testing will be conducted using a 7-inch quadcopter platform, selected to provide a stable and representative test environment for evaluating the flight controller hardware and firmware. This platform offers sufficient payload capacity to accommodate the custom flight controller, external sensors, power distribution hardware, and digital video systems, while remaining manageable for iterative testing and tuning.

The quadcopter configuration will be used to validate core system functionality, including power integrity under dynamic load, data acquisition, actuator signal generation, and manual flight stabilization. Testing on this platform enables controlled evaluation of real-world operating conditions such as vibration, electrical noise from propulsion systems, and environmental disturbances, which are critical for assessing the robustness of the hardware design.

The test platform will be equipped with standard multirotor components, including brushless motors, electronic speed controllers (ESCs), a lithium-polymer battery, and an RC receiver, providing a representative operating environment for the flight controller. Initial validation will focus on bench testing and tethered tests, followed by free flight tests once system stability is confirmed.

If time and system maturity permit, additional validation may be conducted on a fixed-wing or VTOL platform to demonstrate the modularity of the hardware architecture and the flexibility of the sensor and actuator interfaces. Such testing would focus on reuse of the same flight controller hardware with minimal configuration changes, rather than full development of fixed-wing-specific control algorithms. This staged testing approach prioritizes reliable validation of the core hardware platform while allowing optional extension to additional airframe configurations if project constraints are allowed.

4. Expected Outcomes and Deliverables.

Upon completion of the project, the following outcomes are expected:

- A custom-designed six-layer flight controller PCB, including complete schematics, PCB layout, and manufacturing files, developed with explicit consideration for power integrity, signal integrity, and EMI/EMC design principles.
- A fully assembled and electrically validated test board, with verified power rails, clock generation, debug access, and functional communication interfaces.
- Integration and validation of core onboard sensors and peripherals, including inertial measurement units, barometric sensing, CAN communication, and external sensor interfaces, demonstrating reliable data acquisition under bench-test conditions.
- A functional embedded firmware framework, including microcontroller initialization, low-level peripheral drivers, sensor data acquisition, filtering, and basic system services required for flight control operation.
- Implementation of manual flight control logic, including stabilized attitude control and actuator output generation, validated through bench testing and initial flight tests.
- Demonstration of system-level integration, showing correct interaction between hardware, firmware, sensors, actuators, and external modules such as RC receivers and video systems.
- Comprehensive project documentation, including design rationale, hardware and firmware descriptions, test procedures, results, and analysis, as well as final presentation materials.

These outcomes collectively demonstrate the successful design and validation of a modular, hardware-focused flight controller platform capable of supporting multiple UAV configurations and future extensions.

5. Limitations

Despite careful planning and engineering-oriented design decisions, the project is subject to several practical limitations that influence the scope and validation depth of the final system.

5.1 Budget Constraints

The project is developed under a strict budget limitation of approximately 800 GEL ($\approx$ 280 USD). This constraint affects component selection, number of prototype iterations, and feature inclusions. As a result, the design prioritizes modularity, reuse of commercially available components, and a single primary hardware prototype, with limited allowance for multiple PCB fabrication or extensive hardware redundancy.

5.2 EMI/EMC Testing Limitations

While the PCB layout and stack-up are explicitly designed with EMI/EMC compliance principles in mind, formal electromagnetic emission and immunity testing cannot be performed within the scope of this project. Certified EMI/EMC validation requires specialized equipment such as spectrum analyzers, EMI receivers, LISNs, and shielded test environments, which are not accessible under the available budget and laboratory resources.

To mitigate this limitation, the project adopts a design-for-compliance approach, relying on established layout practices and pre-compliance analysis methods. Signal integrity, power integrity, and noise coupling are evaluated using simulation and analytical tools, including circuit-level simulation (e.g., SPICE-based tools, Open EMS), interface modeling, and manufacturer-provided reference models. Layout decisions are guided by industry standards and best practices to reduce the likelihood of EMI/EMC issues prior to any formal certification testing.

5.3 Testing Scope and Validation Depth

System validation is primarily limited to bench testing and flight testing on representative airframes, rather than exhaustive environmental or regulatory testing. As a result, performance validation focuses on functional correctness, stability, and robustness under realistic operating conditions rather than compliance with formal certification thresholds.

These limitations are acknowledged as inherent to an academic, prototype-level project and do not detract from the technical validity of the design. Instead, they reflect realistic engineering trade-offs and reinforce the project's emphasis on sound architecture, standards-aware design, and future extensibility toward more comprehensive testing if resources permit.